



WirelessHART—an extension of the highway addressable remote transducer protocol

connect directly with a central computing device. In most cases, however, sensor nodes communicate in ‘hops’—the message hops from one node to the other before reaching the leader.

Another critical part of the wireless sensor network is the standards and specifications that serve as a guideline to set up the network. “There are various standards and specifications for a variety of sensor applications. For example, Freescale’s ZigBee is a proprietary mesh-networking specification used for embedded sensing, medical data collection, consumer devices (television remote controls) and home automation, while WirelessHART is an extension of the highway addressable remote transducer protocol and specifically designed for industrial applications like process monitoring and control,” says Tyagi.

In addition to the radio transceiver or other wireless communication system, a sensor node contains a small microcontroller and a power source. Some of today’s wireless sensors go a step further and charge themselves using power gleaned from the wireless signals or from ambient sources such as the sun or wind, thereby not requiring a separate power source. This not only increases the scope of applications but also makes the use of sensors practically maintenance-free.

In short, wireless sensors and related technologies—sensor nodes, communication and networking technologies, and software applications that make use of the information collected by the sensors—have all developed significantly in the past few years, extending the potential and practical uses of sensors beyond the mundane.

The following snippets provide a sample of the kind of developments

happening in the wireless sensors space—technological developments, innovative uses and ingenious adaptations!

Quick and easy, green sensor network

At the American Geophysical Union meet held in San Francisco in December last year, a company called Crossbow Solutions demonstrated its award-winning eKo system—a simple, starter-level sensor network kit for gathering environmental data.

The kit consists of sensors, which, in turn, are connected to nodes that read the sensor data and wirelessly transmit it to the base radio. The nodes can send their own information as well as re-transmit data sent from other nodes. The data sent from all nodes is received at the base radio and forwarded to a gateway. The gateway then stores and forwards data from the sensor network, making it accessible via a simple Web-based graphical user interface. While the starter kit has only the basic sensors, environmentalists can expand this array by adding other sensors to precisely measure the parameters they want—be it soil moisture, leaf wetness or the weather.

What makes the eKo system worth a dekho is that the nodes are all solar-powered—making it a green technol-

ogy. The data can also be viewed anywhere, anytime over the Internet, making it easy for even non-techie environmentalists to use. The alarm settings and alerts can be customised, as can be the assortment of sensors used. This makes the system suitable for hobbyists as well as critical environment-monitoring purposes.

In addition to a reliable wireless mesh communication backbone, the system features the environmental sensor bus (ESB) for plug-and-play sensor capability.

Web 2.0 intelligence expanded

Libelium is a company that offers a range of wireless sensor network solutions for requirements ranging from agriculture and cattle farming to healthcare, marketing, industrial processes, logistics and smart metering. The base products comprising their solutions include the Waspnote, a low-power sensor device capable of withstanding adverse environmental conditions, and Meshium, a multi-tech router that enables connections to ZigBee, Wi-Fi, GPRS and Bluetooth networks in addition to GPS geo-location.

Recently, the company released notes on how to send the data collected by its sensor network to Twitter and Wordpress. Basically, it enables the environment to ‘talk’ through a Web presence, updating information about itself in real time.

The Waspnote sensor platform collects the information, while the Meshium router acts as a bridge between the 802.15.4/ZigBee network and the Internet. This open source solution would make it possible to create a blog or micro-blog for a city, forest or a river that updates itself with information collected by the installed sensors—be it about the temperature of a city, the speed or volume of water flowing through a river, the humidity prevalent in a tropical forest or the movement of exotic animals in the forest.

Some months ago, Libelium also released notes on how to program the